



Effect of gonadotropin-releasing hormone administration at the supposed time of artificial insemination on luteinizing hormone and prostaglandin F_{2α} metabolite profiles in lactating Holstein cows

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ABSTRACT

The objective of this study was to evaluate the association between GnRH administration at the supposed time of artificial insemination (AI) and circulating concentrations of LH in lactating Holstein cows, and to explore whether the intensity of estrous expression modulates this relationship. Additionally, we aimed to evaluate the effect of GnRH administration on the uterine response to produce 13,14-dihydro-15-keto-prostaglandin F_{2α} (PGFM) on d 16 of the estrous cycle and explore the association of treatment on progesterone (P4) concentrations following ovulation. A total of 42 lactating cows were submitted to an Ovsynch synchronization protocol. Ovarian ultrasound (US) and blood samples (BS) were assessed at 7 and 14 d after synchronization, and daily until spontaneous estrus event was detected. At the detection of spontaneous estrus, cows were assigned to receive either 100 mg of GnRH (treatment) or equivalent volume of 0.9% sodium chloride solution (control) at the supposed time of AI. Blood samples were collected immediately before treatment administration (0 h) and at 1-h intervals for the following 6 h to evaluate LH concentrations. Subsequent US exams were conducted at 24 h, 48 h, 7 d, and 14 d to confirm ovulation and assess corpus luteum (CL) development. The CL and CL cavity diameter were used to calculate the total CL volume, and BS were collected to measure plasma P4 at 4, 7, 10, and 14 d after estrus. A subset of cows (GnRH: n = 14; control: n = 14) were subjected to an estradiol-oxytocin challenge 16 d after estrus to assess PGFM release. Treatment influenced LH concentrations; however, this effect

was dependent on the initial LH concentration. Cows were categorized by the median LH concentration at 0 h (0.81 ng/mL) into high- or low-LH groups. In high-LH cows, GnRH-treated had greater LH concentrations at 1 h (GnRH: 4.6 ± 0.3 ng/mL; control: 2.3 ± 0.3 ng/mL) and 2 h (GnRH: 3.0 ± 0.3 ng/mL; control: 1.8 ± 0.3 ng/mL) compared with control cows. Estrous intensity did not modulate the LH response, but there was an interaction effect between treatment and intensity of estrus on CL volume and P4 concentrations after estrus. This association was more evident at 14 d after estrus, where control cows with lesser estrous intensity had smaller CL volume and lower P4 concentration than cows with greater estrous intensity and GnRH-treated cows with lesser intensity of estrus (GnRH-greater: 5.0 ± 0.3 ng/mL; control-greater: 5.4 ± 0.3 ng/mL; GnRH-lesser: 5.2 ± 0.3 ng/mL; control-lesser: 4.1 ± 0.3 ng/mL). The GnRH-treated cows also tended to have greater PGFM concentrations compared with control (GnRH: 110.7 ± 9.2 pg/mL; control: 86 ± 9.2 pg/mL). In conclusion, GnRH administration at the supposed time of AI increased circulating LH concentration, although it remains unclear whether this increase is sufficient to influence fertility outcomes. Moreover, estrous intensity appeared to modulate the effect of GnRH on CL development and P4 concentration after estrus.

Key words: GnRH, LH, estrous behavior, dairy cow

INTRODUCTION

Automated activity monitors (AAM) could be a sustainable option for dairy farms to attain the reproductive performance needed to maintain herd profitability while reducing the use of hormones for breeding (Saint-Dizier and Chastant-Maillard, 2012; Gonzalez et al., 2023; Chebel et al., 2025). Several studies have shown that cows displaying spontaneous estrous behavior (Burnett

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et al., 2018; Tippenhauer et al., 2021) or synchronized through hormonal interventions (Galvão et al., 2004; Pereira et al., 2016; Madureira et al., 2019), have increased pregnancy outcomes. In general, not only the occurrence of estrus but also the intensity of estrous expression measured by AAM have been positively associated with pregnancy per artificial insemination (AI), ovulation rates, embryo viability, and reduced pregnancy loss (Burnett et al., 2018; Madureira et al., 2019, 2020). Thus, data generated from AAM could be implemented to identify animals at a greater risk of not becoming pregnant, allowing for targeted interventions and minimizing the use and cost of hormone protocols (Borchardt et al., 2021).

Administration of GnRH at the time of AI in cows with spontaneous estrus is commonly practiced on dairy farms to improve pregnancy outcomes. Exogenous GnRH is well documented to stimulate the anterior pituitary to release LH (Rosenberg et al., 1991; Ortmann et al., 2002), even under elevated progesterone (P4) concentrations (Bolt et al., 1990; Giordano et al., 2012), which could improve ovulation rates and the likelihood of fertilization, thereby explaining observed improvements in fertility. However, its benefit may not extend to all cows, as recent studies suggest that pregnancy outcomes after GnRH treatment at AI may depend on the intensity of estrous expression measured by AAM. In fact, Burnett et al. (2022) demonstrated that GnRH administration at AI improved pregnancy rates in cows displaying low intensity of estrous expression. Yet, although these results may indicate the potential combined use of AAM data with GnRH for targeted reproductive interventions, other studies have reported inconsistent outcomes. Recently, Hubner et al. (2022) reported no effect of GnRH administered at AI on pregnancy per AI in intensity of estrous detected by an AAM. This discrepancy highlights the need for further research on the interaction between GnRH administration at AI and estrous intensity, as well as on the underlying physiological mechanisms involved. Understanding this relationship may help guide on-farm management decisions and improve reproductive outcomes on modern dairy farms.

Administration of GnRH at AI could improve fertility in cows with low intensity of estrous expression by optimizing ovulation timing and increasing the likelihood of fertilization, as these animals have higher ovulation failure rates (Burnett et al., 2018). However, the study by Burnett et al. (2022) observed no differences in ovulation timing. Still, it is possible that exogenous GnRH induces an additional release in LH that could be playing a role not only in the time of ovulation, but also in the luteal cell differentiation and P4 concentrations following ovulation, which would explain fertility outcomes (Hassanein et al., 2024a). Perhaps this additional LH through

exogenous GnRH administration enhances luteal cell differentiation and function following ovulation, improving P4 production, thus favoring uterine receptivity and embryo development (Hassanein et al., 2024b).

Furthermore, it is known that both P4 and estradiol are necessary to synthesize PGF_{2α}, and that the modulation of their concentrations during diestrus can induce a time response, as well as a concentration response, in the production of PGF_{2α} (Schams and Berisha, 2004). Thus, it can be speculated that GnRH administration may also have a downstream effect, likely mediated through P4, on endometrial cells and P4 receptor sensitivity. These effects could affect regulation of the luteolytic cascade, which may affect the uterine capacity to produce PGF_{2α} (Lamming and Mann, 1995). However, this remains hypothetical, as the association among these factors, particularly cows receiving exogenous GnRH, have yet to be investigated.

The main objective of this study was to evaluate the association between GnRH administration at the supposed time of AI and circulating concentrations of LH in lactating Holstein cows, and to explore whether the intensity of estrous expression modulates this relationship. Additionally, we aimed to evaluate the effect of GnRH administration on the uterine response to produce 13,14-dihydro-15-keto-PGF_{2α} (PGFM) on d 16 of the estrous cycle and explore the association of treatment administration on P4 concentrations following ovulation. We hypothesized that lactating dairy cows treated with GnRH would have greater LH concentrations and the intensity of estrus would be associated with the effect of GnRH on LH production. In addition, we hypothesized that GnRH treatment would have an effect on PGFM secretion following the estradiol-oxytocin challenge and concentrations of P4.

MATERIALS AND METHODS

This randomized controlled trial was conducted at the University of British Columbia's Dairy Education and Research Centre in Agassiz, British Columbia, Canada between August and December of 2021. The University of British Columbia's Animal Care Committee approved all procedures in this experiment following protocol A20-0196.

Animals and Housing

A total of 63 lactating Holstein cows (15 primiparous; 48 multiparous) were enrolled in this study. Cows were housed in a wood-framed, freestall barn with deep sand-bedded stalls. Animals were milked twice daily (0500 h and 1500 h) within a conventional milking parlor (Boumatic Dairy Equipment). Fresh TMR, formulated fol-

lowing the *Nutrient Requirements of Dairy Cattle* (NRC, 2001) to meet the requirements of a 720-kg Holstein cow producing 45 kg/d of 3.5% FCM, was fed twice daily (0700 h and 1600 h). Cows had ad libitum access to both TMR and water. Farm personnel were unaware of experimental treatments.

Automated Activity Monitor

Approximately 2 wk before expected calving date, cows were fitted with a leg-mounted AAM (AfiPedometer Plus Tag, AfiMilk) placed on the back leg above the distal expansion of the metatarsal bone. The AAM remained on the cows for the entire lactation, and activity data were recorded hourly throughout the experimental period. The AAM system was checked twice daily (0800 h and 1500 h) for estrus alerts. The peak of activity at estrus was considered as the maximum activity index value (AfiActII, Afimilk) measured by the AAM system. The duration of estrus was defined as the amount of time in hours an animal was above the threshold specified by the AAM. Cows with 2 consecutive estrus alerts within a 4-h interval had the activity data combined to define the total duration of estrus. Cows with 2 consecutive alerts occurring greater than a 5-h interval, the alert with shortest duration was removed from the analysis.

Experimental Design

Farm health records were monitored, and lactating cows between 30 and 60 DIM with no previously recorded health events had their ovarian structures assessed using a portable ultrasound scanner with a 7.5-MHz linear-array transducer (Ibex Pro; E. I. Medical Imaging) to identify the presence of a follicle and corpus luteum (CL). Cows with at least one CL were enrolled in this experiment. The experimental design comprised 4 sequential phases: (1) a synchronization protocol, (2) estrous detection and monitoring until a subsequent spontaneous estrus, (3) detection of the spontaneous estrus followed by treatment, and (4) post-treatment monitoring and estradiol-oxytocin challenge (Figure 1). Farm personnel responsible for daily animal handling were blinded to treatment allocation. Treatment sequence was not completely concealed from members of the research team, however, during ultrasonographic and laboratory evaluations, the individuals conducting the activities did not have direct access to treatment information.

Synchronization Protocol. On d -10, cows received an injection of 100 µg of GnRH (gonadorelin acetate, Fertiline, Vetoquinol) and a 7-d intravaginal P4 implant (1.38 g of progesterone, Zoetis). On d -3, the P4 implant was removed, and an injection of 500 µg of cloprostenol (PGF_{2α}; 2.0 mL i.m.; Estrumate, Merck Animal Health)

was given. After 56 h from implant removal a GnRH injection was administered.

Estrous Monitoring Until a Subsequent Spontaneous Estrus. After the synchronization protocol cows had their BCS recorded, and ovulation was confirmed by ovarian ultrasonography (US). Cows with a dominant follicle >12 mm and either absence of a CL or presence of a CL <24 mm on d 0 had their ovarian structures assessed 7 d later. Cows with the presence of a newly formed CL on the same side of the dominant follicle at d 7 were considered as responsive to the synchronization protocol and kept in the experiment. Beginning on d 14 of the study, daily US examinations were conducted until the AAM detected a subsequent spontaneous estrus. Blood samples (BS; ~10 mL) were collected from the medial coccygeal vein or artery into sterile EDTA tubes (Vacutainer; Becton Dickinson, Franklin Lakes, NJ) at the end of the synchronization protocol (d 0) and at 4, 7, 10, and 14 d, and daily from d 14 until the subsequent estrus event.

Detection of the spontaneous estrus followed by treatment: Spontaneous estrus events were detected by the AAM and confirmed using the same criteria described above (follicle >12 mm and either absence of a CL or presence of a CL <24 mm). At the time of spontaneous estrus, cows' parity, DIM, and week average milk production were retrieved from the farm software (DairyComp 305, Valley Agricultural Software) and used to conduct a block randomization computer-generated sequence (RStudio) to assign cows into 2 treatment groups: GnRH (n = 21) or control (n = 21). The GnRH-treated cows received an intramuscular injection of 100 µg of GnRH (gonadorelin acetate, Fertiline, Vetoquinol) at the supposed time of AI, whereas control cows received an intramuscular injection of 2 mL of 0.9% sodium chloride solution at the supposed time of AI. The supposed timing of AI was determined according to the a.m./p.m. rule based on the onset time of estrus determined by the AAM as follows: Cows with an estrus alert within 0101 h and 1300 h received the treatment in the afternoon of the same day (~1600 h) whereas cows with an estrus alert between 1301 to 0100 h received the treatment in the following morning (~0900 h). Blood samples were collected immediately before treatment (0 h) and hourly for 6 h for LH analysis (Figure 1).

Post-treatment monitoring and estradiol-oxytocin challenge: Once the subsequent spontaneous estrus event was detected, cows confirmed in estrus had their BCS recorded and ovarian structures monitored once daily during the first 48 h after estrus onset, at 4 and 7 d after estrus to confirm new CL formation, and at 14 d after estrus. Blood samples were also collected at 4, 7, 10, and 14 d after spontaneous estrus for determination of plasma P4 concentrations. A subset of cows from each treatment group (GnRH: n = 14; control: n = 14), with

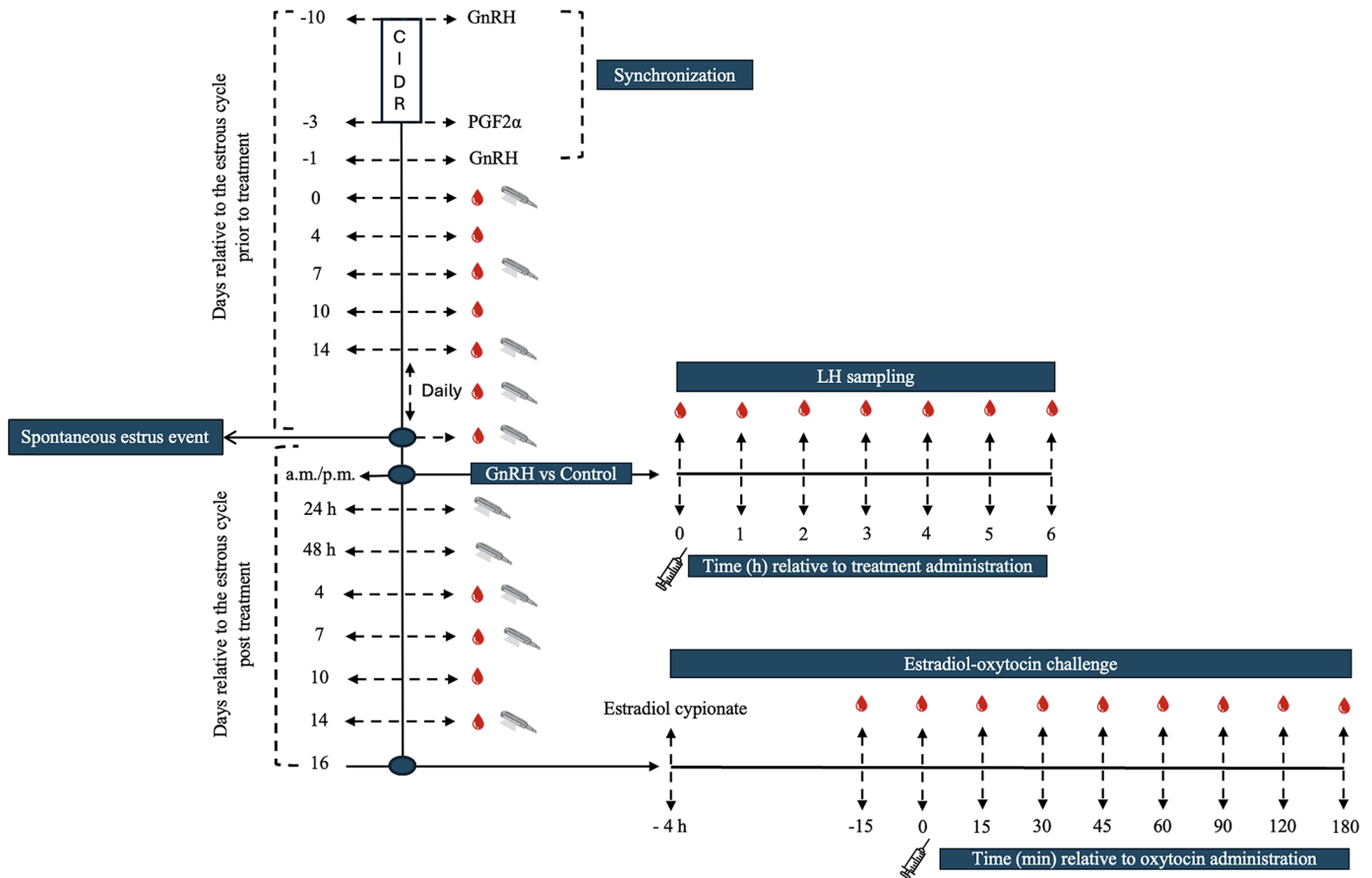


Figure 1. Schematic representation of the synchronization protocol and treatments, along with ultrasound (US) examinations and blood sample (BS) collections performed in the study. Parity, DIM, and milk production were used to randomly assign lactating Holstein cows into 2 treatment groups (GnRH vs. control) to evaluate the effect of GnRH administration at the supposed time of AI on the LH profile and uterine capacity to produce 13,14-dihydro-15-keto-PGF_{2α} (PGFM). The supposed timing of AI was determined according to the a.m./p.m. rule based on the onset time of estrus determined by the AAM as follows: Cows with an estrus alert between 0101 h and 1300 h received the treatment in the afternoon of the same day (~1600 h), whereas cows with an estrus alert between 1301 to 0100 h received the treatment in the following morning (~0900 h). CIDR = controlled internal drug release.

a successful ovulation, were submitted to an estradiol-oxytocin challenge 16 d after the spontaneous estrus event. Cows received an injection of estradiol cypionate (1.0 mg i.m.; Sigma-Aldrich) 4 h before receiving 100 IU of intravenous oxytocin (Oxytocin, Oxyto-sure, Vetoquinol) through the jugular vein. Blood samples were collected 15 min before oxytocin administration, 0 (immediately before oxytocin administration), 15, 30, 45, 60, 90, 120, and 180 min after oxytocin administration for PGFM analysis.

Hormonal Assays

After collection, BS were immediately placed on ice and centrifuged at $2,700 \times g$ for 15 min at 4°C for plasma separation and stored at -20°C until further analysis. Plasma concentration of P4 were measured by an ELISA assay as previously described in Nadalin et al. (2021). The

average intra- and interassay CV was 8.1% and 22.9%, respectively. Plasma concentration of LH was measured by a radioimmunoassay (Bolt and Rollins, 1983; Bolt et al., 1990) with some modifications as outlined in Ginther et al. (1999). Intra-assay CV was 6.2% and sensitivity was 0.05 ng/mL. Additionally, plasma samples from 16 d after estrus were analyzed for PGFM by an ELISA assay (Ginther et al., 2010), with the modifications described by Ochoa et al. (2018). The sensitivity of the assays ranged from 0.1 to 3.7 pg/mL, intra-assay CV ranged from 0.4% to 10% and interassay CV was 14.8%.

Statistical Analysis

A sample size of 15 animals per treatment group was calculated to identify a mean difference of 0.75 ng/mL in LH concentration between treatment groups (GnRH: 2.2 ng/mL vs. Control: 1.45 ng/mL; Stevenson and Pulley,

2016), assuming a variance of 0.3 to 0.4 ng/mL, 95% CI and 80% power. To the best of our knowledge there were no direct studies comparing the association of GnRH at the supposed time of AI and uterine capacity to produce PGFM 16 d after estrus at the time of this experiment. We estimate a sample size of 12 cows per group for PGFM analysis assuming 95% CI and 80% power based on previously reported PGFM concentrations following an estradiol-oxytocin challenge with different P4 concentration treatments (low-P4: 375.5 ± 67.7 pg/mL vs. high-P4: 123.6 ± 55.2 pg/mL; Cerri et al., 2011).

Statistical analyses were performed on SAS Studio (version 3.8, SAS Institute Inc.), and graphical representation was conducted using “ggplot2” package on RStudio (version 2024.09.1, RStudio). Continuous variables were assessed for normality and outliers (PROC UNIVARIATE). Both peak of activity and the duration of estrus were tested separately as covariates in the models due to collinearity, and only peak of activity was used to report the effect of estrous intensity. The median of peak of activity (387 activity-index) at estrus was used to categorize cows as greater or lesser intensity of estrous expression. The volume of CL was calculated using volume formula for a sphere ($4/3\pi r^3$). Volume of the cavity of the CL was subtracted from total CL volume to determine the mass volume. The median of LH at 0 h (0.81 ng/mL) was used to categorize cows into high-LH or low-LH. The supposed time of AI was used to categorize cows into a.m./p.m. treated cows (GnRH-a.m.; control-a.m.; GnRH-p.m.; control-p.m.). Additionally, a sensitivity analysis was conducted excluding animals in the highest quartile of initial LH concentrations to evaluate the effect of GnRH administration independent of the baseline LH imbalance.

The effect of treatment on both LH and PGFM concentrations, were assessed using mixed linear regression models (PROC MIXED) with autoregressive covariance structure and cow as the experimental unit with sample day as a repeated measure. For LH models, the interaction effect of sample time, treatment, and initial LH concentration were evaluated and kept if significant ($P \leq 0.05$). For PGFM models, concentrations of PGFM before oxytocin injection, and the intensity of estrous expression (peak of activity or the duration of estrus) were included as covariates. In addition, the interaction of time and treatment was tested and kept if significant ($P \leq 0.05$). Concentration of P4 at 7 d and 14 d after treatment were also tested for confounding effects on the PGFM models and kept in the model if $P < 0.2$ (Dohoo et al., 2009).

Exploratory analysis for the effect of treatment on CL development and P4 concentration were conducted using mixed linear regression models (PROC MIXED) with autoregressive as the covariance structure, cow as the ex-

perimental unit and sample day as the repeated measure. Categorization of intensity of estrous expression (greater vs. lesser) and the interaction between treatment and intensity of estrus were tested in both P4 and CL models.

Parity, BCS, DIM, ECM, diameter of pre-ovulatory follicle and volume of CL at onset of estrus, P4 concentration at -4, -3, -2, and -1 d relative to spontaneous estrus, and P4 at estrus were tested as fixed effects for confounding on the effect of treatment on LH and PGFM concentration and kept in the model if $P < 0.2$ (Dohoo et al., 2009). All residuals of the mixed models and outcome variables were assessed for normality and homoscedasticity graphically. Results were considered significant if $P \leq 0.05$ and presented as tendencies if had biological significance and $P > 0.05$ and < 0.10 .

RESULTS

Descriptive Statistics

A total of 63 lactating Holstein cows were enrolled into this randomized controlled trial. However, 11 cows failed to ovulate after the presynchronization protocol; 6 cows had the subsequent cycle as a silent estrus (identified by the disappearance of the dominant follicle followed by the formation of a new CL on the same ovary but no alert on the AAM); and 4 cows had a false positive estrous alert detected by the AAM (defined by the presence of a CL greater than 24 mm and absence of a follicle greater than 12 mm through US exam). Therefore, a total of 42 remaining lactating cows were assigned into 2 treatment groups (GnRH: $n = 21$ and control: $n = 21$) and had their LH profiles evaluated. Two cows failed to ovulate following their subsequent estrus and were removed from the analysis evaluating the CL volume and P4 concentrations after treatment. Two cows from the control group had double ovulations following treatment and were kept in the final analysis as no substantial changes in model estimates were observed.

Cow parity, DIM, BCS, and ECM are described in Table 1. There were no differences observed between study treatment groups in the CL diameter during the luteal phase before treatment and at estrus, as well as pre-ovulatory follicle diameter (Table 1; $P > 0.05$). Progesterone concentration 4 d preceding the treatment ($P = 0.55$) and P4 at treatment ($P = 0.67$) was not different between the treatment groups. The peak of activity and the duration of estrus did not differ between treatment groups ($P = 0.87$ and $P = 0.38$, respectively). The concentration of LH at 0 h was greater for cows in the GnRH group compared with cows in the control group (GnRH: 2.7 ± 0.2 ng/mL; control: 1.4 ± 0.2 ng/mL; $P = 0.01$), therefore, LH concentration at 0 h was included as a covariate due to its biological relevance for the interpretation of the study

Table 1. Descriptive cow characteristics (mean $\pm$ SEM) of lactation, DIM, ECM, BCS, and ovarian structures by treatment group¹

Item	GnRH (n = 21)	Control (n = 21)	P-value ²
Lactation (n)	2.6 $\pm$ 0.4	2.6 $\pm$ 0.4	1
DIM at treatment (d)	75.6 $\pm$ 1.5	74.5 $\pm$ 1.7	0.61
ECM closest to treatment (kg)	50.0 $\pm$ 2.6	54.1 $\pm$ 2.3	0.24
BCS at treatment	2.9 $\pm$ 0.06	2.9 $\pm$ 0.04	0.59
LH concentration at 0 h (ng/mL)	2.7 $\pm$ 0.2	1.4 $\pm$ 0.2	0.01
LH concentration at 0 h for high-LH group (ng/mL)	4.1 $\pm$ 0.2	2.9 $\pm$ 0.3	0.01
LH concentration at 0 h for low-LH group (ng/mL)	0.41 $\pm$ 0.03	0.35 $\pm$ 0.02	0.82
Ovulatory follicle diameter at treatment (mm)	21.0 $\pm$ 0.6	21.7 $\pm$ 0.5	0.43
Corpus luteum 4 d before treatment (mm)	28.9 $\pm$ 0.8	29.6 $\pm$ 0.9	0.62
Corpus luteum day of treatment (mm)	19.2 $\pm$ 0.7	18.6 $\pm$ 0.4	0.44
Corpus luteum 7 d after treatment (mm)	30.4 $\pm$ 0.7	29.1 $\pm$ 0.9	0.26
Corpus luteum 14 d after treatment (mm)	33.8 $\pm$ 1.0	32.2 $\pm$ 0.6	0.20

¹Parity, DIM, and milk production were used to randomly assign lactating Holstein cows to receive an intramuscular injection of 100 μ g of GnRH (gonadorelin acetate, Fertiline, Vetoquinol) or the same volume of 0.9% sodium chloride (NaCl) solution at the supposed time of AI.

²Comparisons between treatment groups were conducted using Student's *t*-test. Significance was defined as *P* < 0.05.

findings. The interval from onset of estrus to treatment was on average 8.6 ± 0.3 h (median = 9 h; range = 4–17 h) for GnRH-treated group and 11.2 ± 0.2 h (median = 12 h; range = 4–18 h) for the control group.

Effect of GnRH Administration on LH Profile

There was an effect of initial LH and sample time on LH concentrations (*P* < 0.01). When controlling for initial LH concentration as a continuous covariate, cows that received GnRH had greater LH concentrations at 1 h following treatment (GnRH: 2.9 ± 0.2 ng/mL; control: 1.4 ± 0.2 ng/mL; *P* < 0.01) and tended to have greater LH concentration 2 h after treatment (GnRH: 1.7 ± 0.2 ng/mL; control: 1.2 ± 0.2 ng/mL; *P* = 0.09; Figure 2) compared with the control group. There was no difference in LH concentrations between treatment groups 3 h after treatment regardless of the initial LH.

When cows were categorized based on initial LH concentration into high-LH or low-LH, the effect of treatment was only observed on cows considered as high initial LH. Cows in the high-LH group that received GnRH had greater LH concentration at 1 h (GnRH: 4.6 ± 0.3 ng/mL, *n* = 13; control: 2.3 ± 0.3 ng/mL, *n* = 9; *P* < 0.01; Figure 3) and at 2 h after treatment (GnRH: 3.0 ± 0.3 ng/mL; control: 1.8 ± 0.3 ng/mL; *P* < 0.01; Figure 3) than cows in the control group considered with high initial LH. No differences in LH concentration between treatment groups were observed beyond 2 h. There was no difference between treatment groups on LH concentrations when cows were categorized as having low initial LH concentration (GnRH low-LH, *n* = 8; control low-LH, *n* = 12; *P* = 0.72; Figure 3). Additionally, there was no interaction between the intensity of estrous expression, measured as the peak of activity, and GnRH

treatment on LH concentrations regardless of initial LH concentration (*P* = 0.40).

However, a greater proportion of cows in the GnRH-treated group had higher LH concentration immediately before treatment compared with the control group. When categorizing the groups based on the median of initial LH concentration into high-LH or low-LH, cows considered with high initial LH that received GnRH had greater LH concentration immediately before treatment than cows in the control group considered with high initial LH concentration (GnRH: 4.1 ± 0.2 ng/mL, *n* = 13; control: 2.9 ± 0.3 ng/mL, *n* = 9; *P* < 0.05). Model estimates from the sensitivity analysis excluding animals in the highest quartile of initial LH concentrations were similar to those obtained from the full dataset. Meanwhile, there was no difference in initial LH concentration when cows were considered as low initial LH concentration (GnRH: 0.4 ± 0.03 ng/mL, *n* = 8; control 0.3 ± 0.02 ng/mL, *n* = 12; *P* = 0.13).

Additional analysis of treatment categorized by the supposed time of AI (a.m./p.m.) demonstrated that GnRH-a.m. cows tended to have greater LH concentration 1 h after treatment compared with control-a.m. (GnRH-a.m.: 2.1 ± 0.5 ng/mL, *n* = 8; control-a.m.: 0.91 ± 0.4 ng/mL, *n* = 13; *P* = 0.07), with no differences observed beyond 1 h after treatment (*P* > 0.05). Cows categorized as GnRH-p.m. had greater LH concentration at 1 h (GnRH-p.m.: 3.81 ± 0.4 ng/mL, *n* = 13; control-p.m.: 1.53 ± 0.5 ng/mL, *n* = 8; *P* < 0.001) and at 2 h (GnRH-p.m.: 2.44 ± 0.4 ng/mL, *n* = 13; control-p.m.: 1.17 ± 0.5 ng/mL, *n* = 8; *P* = 0.05) after treatment compared with control-p.m. No differences were observed beyond 2 h after treatment (*P* > 0.05). Yet, independently of the a.m./p.m. categorization, GnRH-treated cows still had numerically greater initial LH concentrations compared with control cows (GnRH-

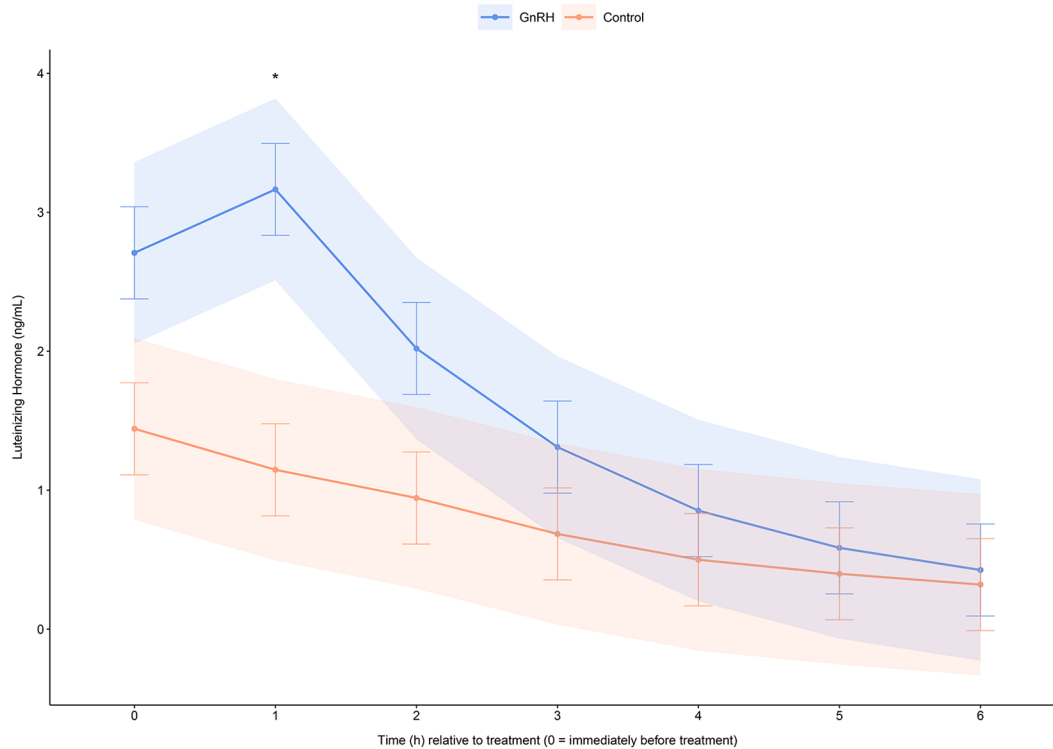


Figure 2. Mean ($\pm$ SEM; shaded area represents 95% CI) LH concentrations (ng/mL) over time (h) by treatment (GnRH or control). Time 0 h corresponds to the sample collected immediately before treatment administration. The * indicates statistical significance ($P < 0.05$).

a.m.: 1.57 ± 0.5 ng/mL; control-AM: 0.94 ± 0.4 ng/mL; $P = 0.33$; and GnRH-p.m.: 3.4 ± 0.4 ng/mL; control-p.m.: 2.26 ± 0.5 ng/mL; $P = 0.08$).

Effect of GnRH Administration on PGFM Profile

Overall concentrations of PGFM assessed following an estradiol-oxytocin challenge tended to be greater for cows in the GnRH group compared with control cows (GnRH: 110.7 ± 9.2 pg/mL; control: 86 ± 9.2 pg/mL; $P = 0.07$; Figure 4). There was an interaction of treatment and sample time on concentrations of PGFM. At 90 min relative to oxytocin administration, cows in the GnRH group had 44 ± 18 pg/mL greater PGFM concentration compared with control cows ($P = 0.02$); all other time points were not different between groups. The intensity of estrus ($P = 0.68$) or P4 concentrations at 14 d after treatment ($P = 0.95$) did not influence the effect of treatment on the PGFM levels following the estradiol-oxytocin challenge.

Exploratory Analysis: Effect of GnRH and Estrous Intensity on Luteal Function

A significant interaction was found between the treatment and the intensity of estrous expression on the CL

volume at 7 and 14 d after treatment. At 7 d, control cows with lesser intensity of estrus had smaller CL volumes than cows with greater intensity of estrous expression regardless of their study treatment (Figure 5; $P < 0.05$). Among lesser intensity estrous cows, there was no difference in the CL volume at 7 d between treatment groups (Figure 5; $P = 0.14$). At 14 d after treatment, however, control cows that displayed lesser intensity of estrus had smaller CL volume than greater intensity of estrus cows and GnRH-treated cows that displayed lesser intensity of estrus (Figure 5; $P < 0.05$).

Similarly, there was a significant interaction between treatment and the intensity of estrous expression on P4 concentrations after treatment. At 7 d, control cows with lesser intensity of estrus tended to have lower circulating concentrations of P4 than GnRH-treated cows that displayed greater intensity of estrus (GnRH-greater: 3.2 ± 0.3 ng/mL; control-lesser: 2.4 ± 0.3 ng/mL; $P = 0.08$; Figure 6). At 10 d, control cows with lesser intensity of estrus had lower P4 concentrations compared with cows displaying greater intensity of estrus (GnRH-greater: 4.6 ± 0.3 ng/mL; control-greater: 4.8 ± 0.3 ng/mL; control-lesser: 3.6 ± 0.3 ng/mL; $P < 0.05$; Figure 6). However, among cows with lesser intensity of estrus, there was no difference at this time in P4 concentration (GnRH-lesser: 4.2 ± 0.3 ng/mL; control-lesser: 3.6 ± 0.3 ng/mL;

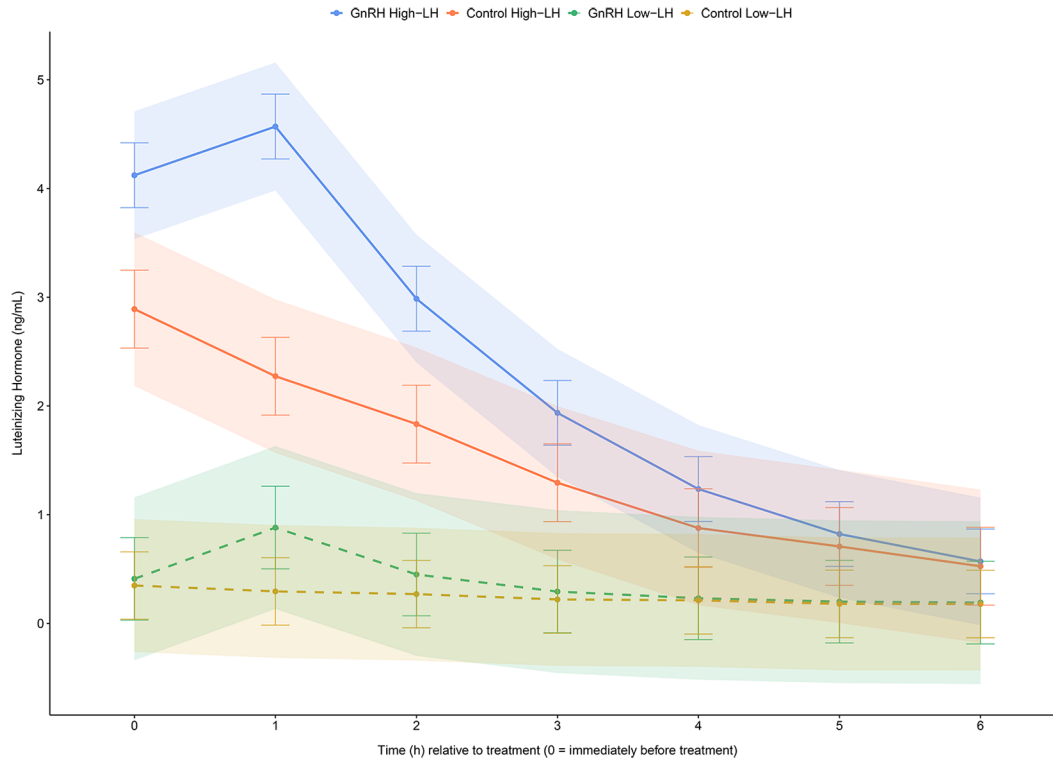


Figure 3. Mean ($\pm$ SEM; shaded area represents 95% CI) LH concentrations (ng/mL) over time (h) by treatment (GnRH or control) and initial LH categorization (high-LH and low-LH). Time 0 h corresponds to the sample collected immediately before treatment administration. Cows were categorized as high-LH or low-LH based on the median LH concentration at 0 h (high-LH: GnRH, $n = 13$; control, $n = 9$; low-LH: GnRH, $n = 8$; control, $n = 12$).

$P = 0.27$; Figure 6). Last, at 14 d after treatment, cows in the control group with lesser intensity of estrus had lower P4 concentrations than greater intensity of estrus cows and lower than GnRH-treated cows that displayed lesser intensity of estrus (GnRH-greater: 5.0 ± 0.3 ng/mL; control-greater: 5.4 ± 0.3 ng/mL; GnRH-lesser: 5.2 ± 0.3 ng/mL; control-lesser: 4.1 ± 0.3 ng/mL; $P < 0.05$; Figure 6). Further animal variation can be observed in the Supplemental Figure S1 (see Notes).

DISCUSSION

The main objective of this study was to evaluate the effect of GnRH administration at the supposed time of AI on the concentrations of LH and to determine whether the intensity of estrous expression modulates this relationship. Overall, our study demonstrated that the administration of GnRH had a small influence on LH concentrations, and that the intensity of estrus did not modulate the response of LH to GnRH. In addition, administration of GnRH only tended to promote greater concentration of PGFM following the estradiol-oxytocin challenge. Perhaps the surprising result from this trial was the positive effect of GnRH on the CL volume and P4 concentrations.

It is known that the LH surge created by exogenous GnRH analogs are not as pronounced as the endogenous one (Ryan et al., 1994; Giordano et al., 2012; Pulley et al., 2015). In this study, the GnRH treatment was administered after the endogenous LH surge, thus the response on LH concentrations was expected to be indeed significantly smaller. Our results demonstrated that initial LH concentration was at nadir levels for most cows, which aligned with reports by Aungier et al. (2015), suggesting that the endogenous LH surge may commence before AAM-detected estrus and persist for about 14 h. Although by chance this study had a greater proportion of cows in the GnRH group with higher LH concentration before treatment than the control group, our results suggest an increase in circulating LH concentrations following GnRH administration even when accounting for initial LH concentrations. However, the effect of GnRH administration was transient and more evident in cows presumably within the endogenous LH surge. This may be attributed to a depletion of the LH reserve in the pituitary following the spontaneous LH surge (Giordano et al., 2012). Arguably, GnRH administered closer to the onset of estrus, concurrent with the endogenous pre-ovulatory LH surge, may result in greater circulating levels of LH

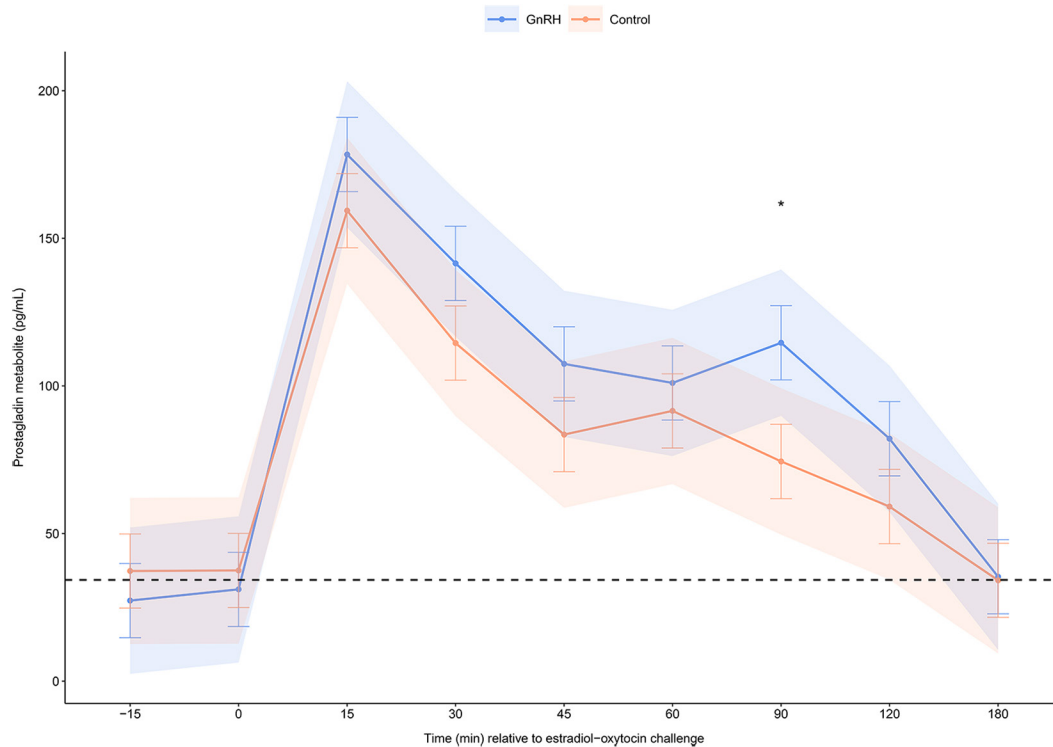


Figure 4. Mean ($\pm$ SEM; shaded area represents 95% CI) concentrations of 13,14-dihydro-15-keto-PGF_{2 α} (PGFM) over time (min) following an estradiol-oxytocin challenge. A subset of cows (GnRH: n = 14 and control: n = 14) were submitted to an estradiol-oxytocin challenge 16 d after estrus. Time 0 corresponds to the sample immediately before oxytocin administration. Dashed line represents mean of PGFM concentrations (34.3 pg/mL) at 0 h. The * indicates statistical significance ($P < 0.05$).

(Kaim et al., 2003; Hassanein et al., 2024a). Additionally, due to the nature of LH release, cows in the control group with elevated initial LH concentration exhibited a natural decrease of LH, indicating that samples were collected at the tail end of the natural LH surge. Whereas cows in the GnRH-treated group showed a slight increased or sustained LH concentration within 1 h after treatment. Nonetheless, we believe that the magnitude of this response (approximately 1.5 ng/mL), compared with the endogenous LH surge variation (8–14 ng/mL; Chenault et al., 1975; Armengol-Gelonch et al., 2017), is likely not biologically relevant and sufficient to enhance downstream reproductive processes and have meaningful effect on fertility outcomes. Given the complex hormonal interplay during the peri-ovulatory period, the observed increase in LH concentration alone may not fully account for the fertility gains observed in other studies (Morgan and Lean, 1993; Roth et al., 2021; Burnett et al., 2022).

Although our findings did not support the initial hypothesis of a modulatory effect of the intensity of estrus on the LH response to GnRH administration, the exploratory analysis showed an interaction between GnRH treatment and estrous expression on CL formation and P4 concentrations after treatment. Administration of

GnRH to cows with lesser intensity of estrus improved circulating levels of P4 in the subsequent cycle to levels comparable to cows with greater intensity of estrus. Particularly, this effect was more pronounced on d 14, which suggests that GnRH administration may benefit luteal development and maintenance instead of initial luteinization. Burnett et al. (2022) reported improved pregnancy per AI with GnRH administration at AI, particularly in cows displaying lesser intensity of estrus. Moreover, previous studies have indicated that relatively small differences in circulating progesterone (~1.5 ng/mL) at d 14 could be associated with pregnancy outcomes in cattle (Larson et al., 1997; Cuevas-Gómez et al., 2025). The observed changes in CL development and concomitant P4 concentrations in the present study may provide insights for the fertility improvements observed in Burnett et al. (2022). However, due to limited sample size and exploratory nature of this analysis, these results should be carefully interpreted and further replication with larger sample size should be conducted to verify the findings observed. Mechanistically, as a meaningful LH surge modulation remains uncertain, a direct effect of GnRH on cellular luteinization could be possible. The presence of GnRH receptors (**GnRHR**) in reproductive

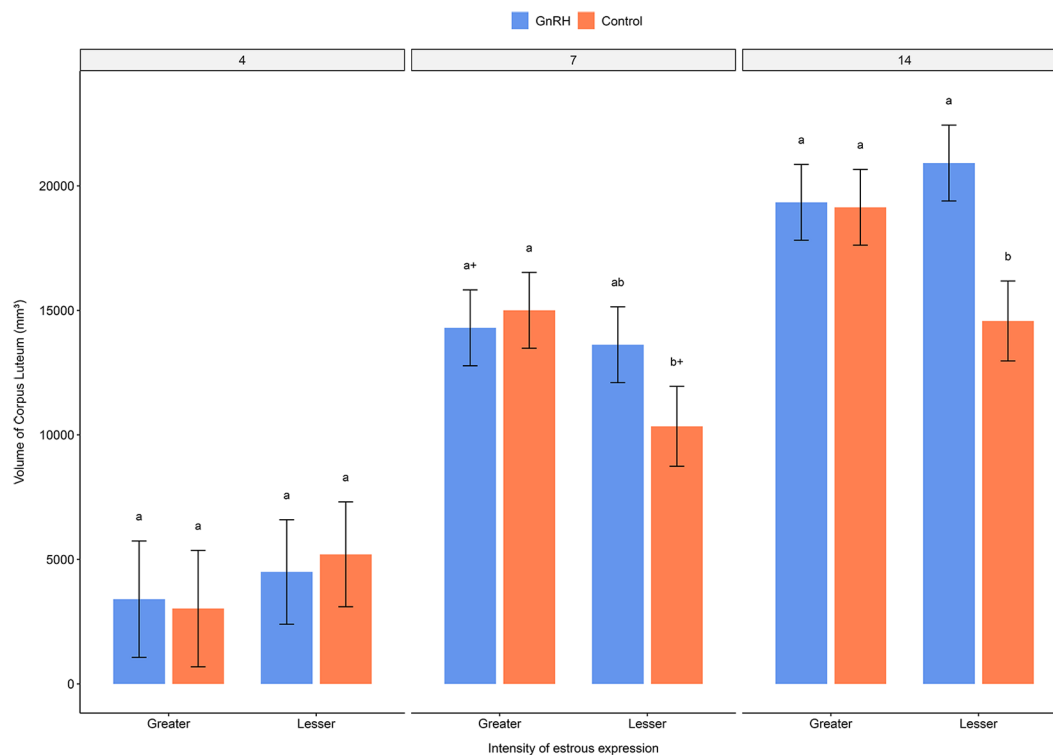


Figure 5. Mean ($\pm$ SEM) of the corpus luteum volume at 4, 7, and 14 d after treatment. Parity, DIM, and milk production were used to randomly assign lactating Holstein cows into 2 treatments (GnRH: $n = 21$ and control: $n = 21$) and retrospectively categorize cows by median peak of activity into greater or lesser estrous intensity (GnRH-greater: $n = 10$, GnRH-lesser: $n = 10$, control-greater: $n = 10$, and control-lesser: $n = 10$). Different letters indicate statistical significance ($P < 0.05$); + indicates statistical tendency ($P > 0.05$ and $P < 0.10$).

tissue beyond the hypothalamus and pituitary have been previously documented (Ramakrishnappa et al., 2005; Metallinou et al., 2007). For instance, GnRHR located on the granulosa cells in the ovary might play a local role by enhancing the enzymatic activity of cytochrome P450 side-chain cleavage or steroid acute regulatory protein (D'Occhio and Aspden, 1999). This increase in activity could facilitate cholesterol mobilization and granulosa cell differentiation into luteal cells, thereby supporting CL development and P4 production in the subsequent cycle (Schams and Berisha, 2004; Hassanein et al., 2024a). This effect might be more pronounced in cows with lesser estrous intensity, which possibly have sub-optimal hormonal profiles (Madureira et al., 2021) and receptor expression in their reproductive tissues. However, direct effects of GnRH on luteal cell differentiation, particularly associating with estrous intensity, have not been investigated and remain speculative. It is unclear whether this mechanism contributes to the observed interaction, and further research with larger sample size is required to evaluate the association between GnRH treatment, estrous intensity and postestrus P4 concentrations. Nonetheless, the existing literature presents contradictory results on fertility outcomes following GnRH

administration and the mechanism of action of GnRH remains unclear (Morgan and Lean, 1993; Valenza et al., 2012; Roth et al., 2021; Hubner et al., 2022). Potentially, there is a sensitive window for the time of GnRH administration relative to onset of estrus that might have an impact on the effect observed of GnRH administration (Rosenberg et al., 1991). Further investigations should explore the intensity of estrus and timing of GnRH following estrus in modern dairy cows.

Cows administered GnRH tended to have greater circulating levels of PGFM following the estradiol-oxytocin challenge. The potential effect of GnRH to enhance luteal cell differentiation (Hassanein et al., 2024a) and therefore, P4 production, may improve uterine receptivity in favor of improved embryo development or may affect the uterine capacity to produce prostaglandin 16 d after estrus (Hassanein et al., 2024b). The loss of sensitivity and downregulation of P4 receptors in endometrium cells allows for the upregulation of estradiol and oxytocin receptors expression in the uterus that would lead to pulses of $\text{PGF}_{2\alpha}$ and activation of luteolytic cascade (Lamming and Mann, 1995). The increase in PGFM observed in our study may be associated with an enhanced endometrial responsiveness following the estradiol-oxytocin chal-

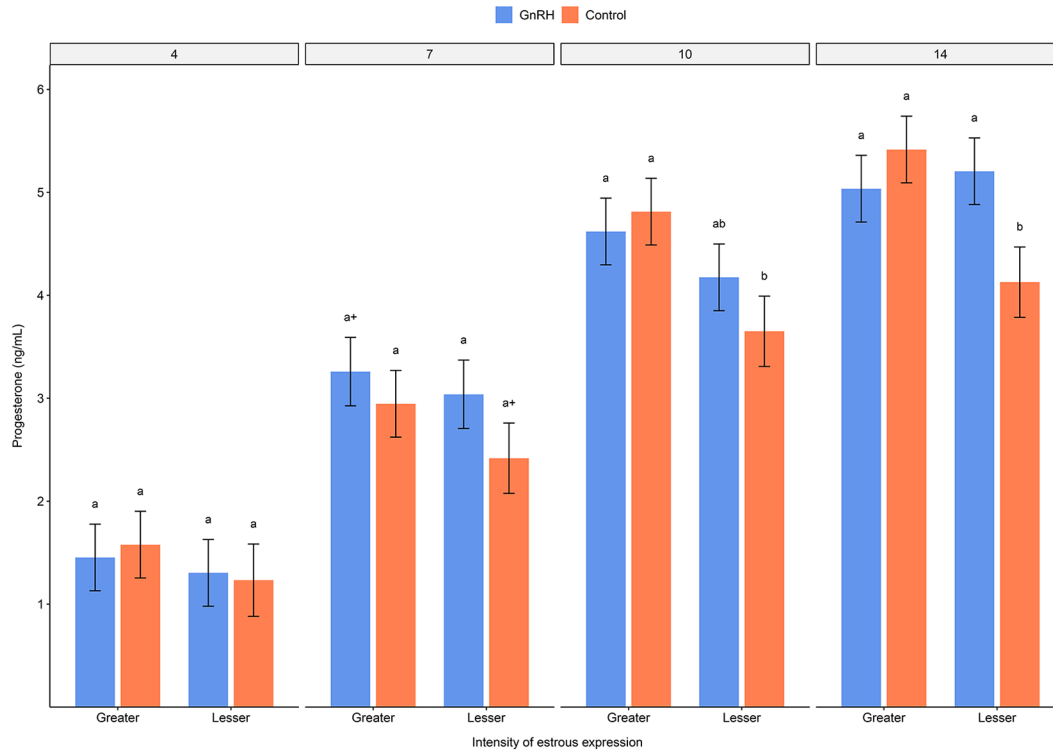


Figure 6. Mean ($\pm$ SEM) of progesterone (P4) concentration at 4, 7, 10, and 14 d after treatment. Parity, DIM, and milk production were used to randomly assign lactating Holstein cows into 2 treatments (GnRH: $n = 21$ and control: $n = 21$) and retrospectively categorize cows by median peak of activity into greater or lesser estrous intensity (GnRH-greater: $n = 10$, GnRH-lesser: $n = 9$, control-greater: $n = 10$, and control-lesser: $n = 10$). Different letters indicate significance ($P < 0.05$); + indicates statistical tendency ($P > 0.05$ and $P < 0.10$).

lenge. Yet, there was no association between P4 at d 14 and PGFM concentrations. It is possible that this lack of association could be attributed to our inability to evaluate the interaction of treatment with estrous expression, due to sample size restrictions, given that treatment alone was not associated with P4 concentration after estrus. Additionally, it is possible that GnRH may have a direct or indirect effect not previously hypothesized in this manuscript. Circulating PGFM concentrations were transient, and PGFM values (GnRH: 110.7 ± 9.2 pg/mL; control: 86 ± 9.2 pg/mL) were comparable to those previously reported by Cerri et al. (2011) in cows exposed to high P4 concentrations and much lower when compared with cows exposed to low P4 treatment (high-P4: 123.6 ± 55.2 pg/mL; low-P4: 375.5 ± 67.7 pg/mL; Cerri et al., 2011). Although significant, this increase is unlikely to be biologically relevant, and further research is needed to elucidate the potential effects of GnRH on the uterine environment. In addition, the estradiol-oxytocin challenge requires careful interpretation as it primarily assesses the endometrial responsiveness to produce PGFM, which does not fully replicate the endogenous pulsatile PGF_{2 α} cascade required for luteolysis activation. A transient increase in circulating PGFM may therefore reflect en-

dometrial responsiveness without indicating premature risk for natural luteolysis or compromising CL lifespan, shorten cycle length, or impair embryo survival.

Although cows were controlled for parity, DIM and milk production, and were assigned to treatment groups as the spontaneous estrus was detected, the time of treatment was predetermined to represent the time of AI (approximately 0900 and 1600 h). A limitation of this study was the unequal distribution of initial LH concentrations between treatment groups. By chance, a greater proportion of cows in the GnRH-treated group had shorter interval from onset of estrus to treatment and higher LH concentration immediately before treatment compared with the control group. These imbalances may have influenced the hormonal responses observed, prompting the decision to include initial LH concentration as a covariate in the analyses and the categorization of animals based on initial LH concentration. The imbalance in initial LH concentration and multiple comparison burden may have limited the association of GnRH administration and estrous expression on LH concentrations. Additionally, treatment allocation was not completely concealed from all team members, which may represent a potential for selection or performance

bias; however, individuals performing ultrasonographic and laboratory evaluations did not have direct access to treatment information, thereby reducing the likelihood of selection bias influencing outcome assessment. Although treatment differences were observed in CL volume and P4 concentrations, the relatively small sample size may increase the likelihood of detecting these differences by chance. Furthermore, this study was conducted within a single herd during a single season, which may limit the generalizability of the results and fertility outcomes, such as pregnancy per AI, were not evaluated; therefore, the physiological responses observed herein cannot be directly linked to reproductive performance.

We encourage future studies to assess the endogenous LH surge to more precisely evaluate the effects of GnRH administration in relation to the timing of onset of estrus detected by AAM. In addition, investigation of GnRHR expression in granulosa and luteal cells of cows exhibiting differing intensities of estrous expression may help elucidate underlying physiological mechanisms. More importantly, further research should investigate the effects of GnRH administration on the recovery of luteal function, particularly P4 profiles beyond 10 d of the estrous cycle, in cows expressing lesser estrous intensity. With further physiological understanding and data prediction, GnRH administration could be integrated with AAM generated data to support targeted interventions in modern dairy farms. To date, although targeted GnRH administration appears to be a feasible approach for lesser intensity of estrus, most studies using AAM data relied on retrospective estrus classification which limits practical application.

CONCLUSIONS

This study demonstrated that administration of GnRH at the supposed time of AI influenced LH concentrations, with the magnitude of the response dependent on cows' initial LH profile. However, the intensity of estrus did not appear to modulate the LH response to GnRH. Administration of GnRH tended to promote a modest increase in PGFM concentrations following the estradiol-oxytocin challenge, though the biological relevance of this effect remains uncertain. Last, our exploratory analysis demonstrated that control cows expressing lesser intensity of estrus had smaller CL volume after treatment and lower P4 concentrations than cows expressing greater intensity of estrus regardless of treatment group.

NOTES

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Nonstandard abbreviations used: AAM = automated activity monitors; AI = artificial insemination; BS = blood sample; CIDR = controlled internal drug release; CL = corpus luteum; GnRHR = GnRH receptor; P4 = progesterone; PGFM = 13,14-dihydro-15-keto-PGF_{2α}; US = ultrasound.

REFERENCES

- Armengol-Gelonch, R., J. M. Mallo, D. Ponté, A. Jimenez, A. Valenza, and A. H. Souza. 2017. Impact of phase of the estrous cycle and season on LH surge profile and fertility in dairy cows treated with different GnRH analogs (gonadorelin vs. buserelin). *Theriogenology* 91:121–126. <https://doi.org/10.1016/j.theriogenology.2017.01.001>.
- Aungier, S. P. M., J. F. Roche, P. Duffy, S. Scully, and M. A. Crowe. 2015. The relationship between activity clusters detected by an automatic activity monitor and endocrine changes during the periestrous period in lactating dairy cows. *J. Dairy Sci.* 98:1666–1684. <https://doi.org/10.3168/jds.2013-7405>.
- Bolt, D. J., and R. Rollins. 1983. Development and application of a radioimmunoassay for bovine follicle-stimulating hormone. *J. Anim. Sci.* 56:146–154. <https://doi.org/10.2527/jas1983.561146x>.
- Bolt, D. J., V. Scott, and G. H. Kiracofe. 1990. Plasma LH and FSH after estradiol, norgestomet and GnRH treatment in ovariectomized beef heifers. *Anim. Reprod. Sci.* 23:263–271. [https://doi.org/10.1016/0378-4320\(90\)90040-M](https://doi.org/10.1016/0378-4320(90)90040-M).
- Borchardt, S., C. M. Tippenhauer, J.-L. Plenio, A. Bartel, A. M. L. Madureira, R. L. A. Cerri, and W. Heuwieser. 2021. Association of estrous expression detected by an automated activity monitoring system within 40 days in milk and reproductive performance of lactating Holstein cows. *J. Dairy Sci.* 104:9195–9204. <https://doi.org/10.3168/jds.2020-19705>.
- Burnett, T. A., A. M. L. Madureira, J. W. Bauer, and R. L. A. Cerri. 2022. Impact of gonadotropin-releasing hormone administration at the time of artificial insemination on conception risk and its association with estrous expression. *J. Dairy Sci.* 105:1743–1753. <https://doi.org/10.3168/jds.2021-20156>.
- Burnett, T. A., L. Polsky, M. Kaur, and R. L. A. Cerri. 2018. Effect of estrous expression on timing and failure of ovulation of Holstein dairy cows using automated activity monitors. *J. Dairy Sci.* 101:11310–11320. <https://doi.org/10.3168/jds.2018-15151>.
- Cerri, R. L. A., R. C. Chebel, F. Rivera, C. D. Narciso, R. A. Oliveira, M. Amstalden, G. M. Baez-Sandoval, L. J. Oliveira, W. W. Thatcher, and J. E. P. Santos. 2011. Concentration of progesterone during the development of the ovulatory follicle: II. Ovarian and uterine responses. *J. Dairy Sci.* 94:3352–3365. <https://doi.org/10.3168/jds.2010-3735>.
- Chebel, R. C., T. Gonzalez, A. B. Montevecchio, K. N. Galvão, A. de Vries, and R. S. Bisinotto. 2025. Targeted reproductive manage-

- ment for lactating Holstein cows: Economic return. *J. Dairy Sci.* 108:1584–1601. <https://doi.org/10.3168/jds.2024-25525>.
- Chenault, J. R., W. W. Thatcher, P. S. Kalra, R. M. Abrams, and C. J. Wilcox. 1975. Transitory changes in plasma progesterone, estradiol, and luteinizing hormone approaching ovulation in the bovine. *J. Dairy Sci.* 58:709–717. [https://doi.org/10.3168/jds.S0022-0302\(75\)84632-7](https://doi.org/10.3168/jds.S0022-0302(75)84632-7).
- Cuevas-Gómez, I., L. Molina, J. F. Sánchez-Madueño, I. Sánchez-Madueño, P. Lonergan, D. Rizos, C. C. Pérez-Marín, and J. M. Sánchez. 2025. Circulating progesterone concentrations and pregnancy outcomes in high-producing lactating dairy cows treated with human chorionic gonadotropin on day 2 of the estrus cycle. *J. Dairy Sci.* 108:5448–5461. <https://doi.org/10.3168/jds.2024-26000>.
- D'Occhio, M. J., and W. J. Aspden. 1999. Endocrine and reproductive responses of male and female cattle to agonists of gonadotrophin-releasing hormone. *J. Reprod. Fertil. Suppl.* 54:101–114.
- Dohoo, I. R., S. W. Martin, and H. Stryhn. 2009. *Veterinary Epidemiologic Research*. 2nd ed. VER Inc.
- Galvão, K. N., J. E. P. Santos, S. O. Juchem, R. L. A. Cerri, A. C. Cosciani, and M. Villaseñor. 2004. Effect of addition of a progesterone intravaginal insert to a timed insemination protocol using estradiol cypionate on ovulation rate, pregnancy rate, and late embryonic loss in lactating dairy cows. *J. Anim. Sci.* 82:3508–3517. <https://doi.org/10.2527/2004.82123508x>.
- Ginther, O. J., D. R. Bergfelt, L. J. Kulick, and K. Kot. 1999. Selection of the dominant follicle in cattle: establishment of follicle deviation in less than 8 hours through depression of FSH concentrations. *Theriogenology* 52:1079–1093. [https://doi.org/10.1016/S0093-691X\(99\)00196-X](https://doi.org/10.1016/S0093-691X(99)00196-X).
- Ginther, O. J., H. K. Shrestha, M. J. Fuenzalida, A. K. M. Shahiduzzaman, and M. A. Beg. 2010. Characteristics of pulses of 13,14-dihydro-15-keto-prostaglandin F₂α before, during, and after spontaneous luteolysis and temporal intrapulse relationships with progesterone concentrations in cattle. *Biol. Reprod.* 82:1049–1056. <https://doi.org/10.1095/biolreprod.109.081976>.
- Giordano, J. O., P. M. Fricke, J. N. Guenther, G. Lopes Jr., M. M. Herlihy, A. B. Nascimento, and M. C. Wiltbank. 2012. Effect of progesterone on magnitude of the luteinizing hormone surge induced by two different doses of gonadotropin-releasing hormone in lactating dairy cows. *J. Dairy Sci.* 95:3781–3793. <https://doi.org/10.3168/jds.2011-5155>.
- Gonzalez, T. D., L. Factor, A. Mirzaei, A. B. Montevecchio, S. Casaro, V. R. Merenda, J. G. Prim, K. N. Galvão, R. S. Bisinotto, and R. C. Chebel. 2023. Targeted reproductive management for lactating Holstein cows: Reducing the reliance on exogenous reproductive hormones. *J. Dairy Sci.* 106:5788–5804. <https://doi.org/10.3168/jds.2022-22666>.
- Hassanein, E. M., Z. Szelényi, and O. Szenci. 2024a. Gonadotropin-releasing hormone (GnRH) and its agonists in bovine reproduction I: Structure, biosynthesis, physiological effects, and its role in estrous synchronization. *Animals (Basel)* 14:1473. <https://doi.org/10.3390/ani14101473>.
- Hassanein, E. M., Z. Szelényi, and O. Szenci. 2024b. Gonadotropin-releasing hormone (GnRH) and its agonists in bovine reproduction II: Diverse applications during insemination, post-insemination, pregnancy, and postpartum periods. *Animals (Basel)* 14:1575. <https://doi.org/10.3390/ani14111575>.
- Hubner, A. M., I. F. Canisso, P. M. Peixoto, A. J. Conley, and F. S. Lima. 2022. Effect of gonadotropin-releasing hormone administered at the time of artificial insemination for cows detected in estrus by conventional estrus detection or an automated activity-monitoring system. *J. Dairy Sci.* 105:831–841. <https://doi.org/10.3168/jds.2021-21011>.
- Kaim, M., A. Bloch, D. Wolfenson, R. Braw-Tal, M. Rosenberg, H. Voet, and Y. Folman. 2003. Effects of GnRH administered to cows at the onset of estrus on timing of ovulation, endocrine responses, and conception. *J. Dairy Sci.* 86:2012–2021. [https://doi.org/10.3168/jds.S0022-0302\(03\)73790-4](https://doi.org/10.3168/jds.S0022-0302(03)73790-4).
- Lamming, G. E., and G. E. Mann. 1995. Control of endometrial oxytocin receptors and prostaglandin F₂α production in cows by progesterone and oestradiol. <https://doi.org/10.1530/jrf.0.1030069>.
- Larson, S. F., W. R. Butler, and W. B. Currie. 1997. Reduced fertility associated with low progesterone postbreeding and increased milk urea nitrogen in lactating cows. *J. Dairy Sci.* 80:1288–1295. [https://doi.org/10.3168/jds.S0022-0302\(97\)76058-2](https://doi.org/10.3168/jds.S0022-0302(97)76058-2).
- Madureira, A. M. L., T. A. Burnett, S. Borchardt, W. Heuwieser, C. F. Baes, J. L. M. Vasconcelos, and R. L. A. Cerri. 2021. Plasma concentrations of progesterone in the preceding estrous cycle are associated with the intensity of estrus and fertility of Holstein cows. *PLoS One* 16:e0248453. <https://doi.org/10.1371/journal.pone.0248453>.
- Madureira, A. M. L., T. A. Burnett, K. G. Pohler, T. G. Guida, C. P. Sanches Jr., J. L. M. Vasconcelos, and R. L. A. Cerri. 2020. Short communication: Greater intensity of estrous expression is associated with improved embryo viability from superovulated Holstein heifers. *J. Dairy Sci.* 103:5641–5646. <https://doi.org/10.3168/jds.2019-17772>.
- Madureira, A. M. L., L. B. Polsky, T. A. Burnett, B. F. Silper, S. Soriano, A. F. Sica, K. G. Pohler, J. L. M. Vasconcelos, and R. L. A. Cerri. 2019. Intensity of estrus following an estradiol-progesterone-based ovulation synchronization protocol influences fertility outcomes. *J. Dairy Sci.* 102:3598–3608. <https://doi.org/10.3168/jds.2018-15129>.
- Metallinou, C., B. Asimakopoulou, A. Schröer, and N. Nikolettos. 2007. Gonadotropin-releasing hormone in the ovary. *Reprod. Sci.* 14:737–749. <https://doi.org/10.1177/1933719107310707>.
- Morgan, W. F., and I. J. Lean. 1993. Gonadotrophin-releasing hormone treatment in cattle: a meta-analysis of the effects on conception at the time of insemination. *Aust. Vet. J.* 70:205–209. <https://doi.org/10.1111/j.1751-0813.1993.tb03304.x>.
- Nadalin, A., J. Denis-Robichaud, A. M. L. Madureira, T. A. Burnett, J. Bauer, J. L. M. Vasconcelos, K. G. Pohler, A. M. Crespihlo, and R. L. A. Cerri. 2021. Technical note: Validation of an in-house bovine serum enzyme immunoassay for progesterone measurement. *J. Dairy Sci.* 104:2455–2462. <https://doi.org/10.3168/jds.2020-18824>.
- NRC. 2001. *Nutrient Requirements of Dairy Cattle: Seventh Revised Edition, 2001*. The National Academies Press, Washington, DC.
- Ochoa, J. C., F. Peñagaricano, G. M. Baez, L. F. Melo, J. C. L. Motta, A. Garcia-Guerra, R. Meidan, J. C. Pinheiro Ferreira, R. Sartori, and M. C. Wiltbank. 2018. Mechanisms for rescue of corpus luteum during pregnancy: gene expression in bovine corpus luteum following intrauterine pulses of prostaglandins E1 and F₂α. *Biol. Reprod.* 98:465–479. <https://doi.org/10.1093/biolre/iox183>.
- Ortmann, O., J. M. Weiss, and K. Diedrich. 2002. Gonadotrophin-releasing hormone (GnRH) and GnRH agonists: Mechanisms of action. *Reprod. Biomed. Online* 5(Suppl 1):1–7. [https://doi.org/10.1016/S1472-6483\(11\)60210-1](https://doi.org/10.1016/S1472-6483(11)60210-1).
- Pereira, M. H. C., M. C. Wiltbank, and J. L. M. Vasconcelos. 2016. Expression of estrus improves fertility and decreases pregnancy losses in lactating dairy cows that receive artificial insemination or embryo transfer. *J. Dairy Sci.* 99:2237–2247. <https://doi.org/10.3168/jds.2015-9903>.
- Pulley, S. L., D. H. Keisler, and J. S. Stevenson. 2015. Concentrations of luteinizing hormone and ovulatory responses in dairy cows before timed artificial insemination. *J. Dairy Sci.* 98:6188–6201. <https://doi.org/10.3168/jds.2015-9473>.
- Ramakrishnapa, N., R. Rajamahendran, Y.-M. Lin, and P. C. K. Leung. 2005. GnRH in non-hypothalamic reproductive tissues. *Anim. Reprod. Sci.* 88:95–113. <https://doi.org/10.1016/j.anireprosci.2005.05.009>.
- Rosenberg, M., S. Y. Chun, M. Kaim, Z. Herz, and Y. Folman. 1991. The effect of GnRH administered to dairy cows during oestrus on plasma LH and conception in relation to the time of treatment and insemination. *Anim. Reprod. Sci.* 24:13–24. [https://doi.org/10.1016/0378-4320\(91\)90078-E](https://doi.org/10.1016/0378-4320(91)90078-E).
- Roth, Z., Y. Z. Kressel, Y. Lavon, D. Kalo, and D. Wolfenson. 2021. Administration of GnRH at onset of estrus, determined by automatic

- activity monitoring, to improve dairy cow fertility during the summer and autumn. *Animals (Basel)* 11:2194. <https://doi.org/10.3390/ani11082194>.
- Ryan, D. P., S. Snijders, T. Condon, M. Grealy, J. Sreenan, and K. J. O'Farrell. 1994. Endocrine and ovarian responses and pregnancy rates in dairy cows following the administration of a gonadotrophin releasing hormone analog at the time of artificial insemination or at mid-cycle post insemination. *Anim. Reprod. Sci.* 34:179–191. [https://doi.org/10.1016/0378-4320\(94\)90015-9](https://doi.org/10.1016/0378-4320(94)90015-9).
- Saint-Dizier, M., and S. Chastant-Maillard. 2012. Towards an automated detection of oestrus in dairy cattle. *Reprod. Domest. Anim.* 47:1056–1061. <https://doi.org/10.1111/j.1439-0531.2011.01971.x>.
- Schams, D., and B. Berisha. 2004. Regulation of corpus luteum function in cattle—An overview. *Reprod. Domest. Anim.* 39:241–251. <https://doi.org/10.1111/j.1439-0531.2004.00509.x>.
- Stevenson, J. S., and S. L. Pulley. 2016. Feedback effects of estradiol and progesterone on ovulation and fertility of dairy cows after gonadotropin-releasing hormone-induced release of luteinizing hormone. *J. Dairy Sci.* 99:3003–3015. <https://doi.org/10.3168/jds.2015-10091>.
- Tippenhauer, C. M., J. L. Plenio, A. M. L. Madureira, R. L. A. Cerri, W. Heuwieser, and S. Borchardt. 2021. Factors associated with estrous expression and subsequent fertility in lactating dairy cows using automated activity monitoring. *J. Dairy Sci.* 104:6267–6282. <https://doi.org/10.3168/jds.2020-19578>.
- Valenza, A., J. O. Giordano, G. Lopes Jr., L. Vincenti, M. C. Amundson, and P. M. Fricke. 2012. Assessment of an accelerometer system for detection of estrus and treatment with gonadotropin-releasing hormone at the time of insemination in lactating dairy cows. *J. Dairy Sci.* 95:7115–7127. <https://doi.org/10.3168/jds.2012-5639>.